AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM298
Operating Systems

Final Examination
Spring Semester 2024/2025
Date: 12/June/2025

Number of Exam Pages: 4 (including this cover sheet)	**Time Allowed:** 2 Hours

Instructions:

- Total Marks:**50**
- This exam consists of **three** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Programmable Calculators** are **not** allowed.

AOU EXAM PAPER

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Part 1:Multiple Choice Questions: [10 marks]

Question 1: Choose the most correct answer for each of the following:

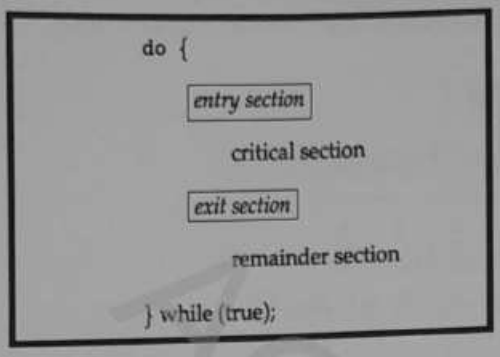
1. Which mode allows the OS to protect itself from user processes?
   a) User mode
   b) Kernel mode
   c) User and kernel modes
   d) Application mode.
2. Which condition is not necessary for the deadlock to occur?
   a) Mutual Exclusion.
   b) Resource Sharing.
   c) Hold and wait.
   d) None of the choices
3. In ____multithreading model, creating a user-level thread creates a kernel thread:
   a) One-to-One
   b) Many-to-One
   c) One-to-Many
   d) Many-to-Many.
4. The correct order of operations for protecting a critical section using **Mutex** locks is:
   a) signal() followed by wait()
   b) release() followed by acquire()
   c) wait() followed by signal()
   d) acquire() followed by release()
5. ___ assigns final addresses to program parts and adjusts code and data in program to match those addresses:
   a) Linker.
   b) Loader.
   c) Object.
   d) Relocation.
6. Which of the following is a disadvantage of Priority Scheduling?
   a) Inefficient CPU utilization.
   b) Requires complex hardware
   c) May lead to starvation.
   d) Simple to implement.
7. Device controller informs CPU that it has finished its operation by causing a/an:
   a) Request
   b) Semaphore
   c) Interrupt
   d) Mutex handling
8. ____ can be used to prevent busy waiting when implementing a semaphore.
   a) Spinlocks
   b) Allowing the wait() operation to succeed.
   c) Mutex lock
   d) Waiting queues
9. A significant problem with priority scheduling algorithms is:
   a) Deadlock
   b) Starvation
   c) Complexity
   d) All the choices
10. _____ is the number of processes that are completed per time unit:
    a) Response time.
    b) Wait time
    c) Throughput.
    d) CPU utilization

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Part 2: Short and Essay questions: [15 marks]

Question 2: [3 marks]
Name two main goals of process scheduling in an operating system.

Question 3: [4 marks]
Given the general structure to deal with the critical section.
Answer the following questions:



a- What is the critical section? (2 marks)
b- What does the exit section do? (2 marks)

Question 4: [4 marks]
Discuss how deadlocks can be prevented and list the necessary conditions for a deadlock to happen.

Question 5: [4 marks]
Two operations are used to avoid busy waiting in a semaphore implementation- block and wakeup. Discuss these two operations.

Part 3: Problem Solving questions: [25 marks]

Question 6: [6 marks]
Consider the following segment table:

Segment	Base	Limit
0	260	175
1	420	100
2	150	120
3	600	180
4	800	400

Find the physical addresses for the following logical addresses and deduce whether this address will be of legal reference or not. (Show the calculations)

a) Segment: 0, Logical address: 175 (2 marks)
b) Segment: 1, Logical address: 80 (2 marks)
c) Segment: 3, Logical address: 150 (2 marks)

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Question 7: [6 marks]

Consider a system with the given main memory, page, and process specifications:

Main Memory size: **4096 KB**; Page size: **64 KB**

- **Process P1**: 1024 KB
- **Process P2**: 880 KB
- **Process P3**: 1232 KB

a- How many **pages** are required for each process? (3 marks)

b- How many **frames** will there be in the main memory? (1 mark)

c- Will the main memory be able to accommodate the three processes to be loaded into it? Discuss. (2 marks)

Question 8: [6 marks]

Consider the following page reference string:

Page reference string: **9,5,2,3,5,4,6,4,3,9**

Ref String	9	5	2	3	5	4	6	4	3	9
f1										
f2										
f3										
H/PF										

Assume that there are **three frames**, and the **LRU** was used. Answer the following questions:

a- Fill in the table above after copying it to your answer booklet. (5 marks)

b- Find the number of Page Faults (PF). (1 mark)

N.B: in the last row, you should indicate whether we have a hit (**H**) or a page fault (**PF**)

Question 9: [7 marks]

Assume that we have the following **Scenario:**

There are four resources (R1, R2, R3, R4) with the following instances:

- R1: One instance
- R2: Three instances
- R3: Two instances
- R4: One instance

There are three processes (P1, P2, P3) with the following resource allocations and requests:

- P1: Holds one instance of R3 and is waiting for one instance of R1.
- P2: Holds one instance of R1 and one instance of R4 and is waiting for one instance of R3.
- P3: Holds one instance of R2 and one instance of R3 and is waiting for one instance of R4.

a) Draw the corresponding Resource-Allocation Graph (RAG). (4 marks)

b) Determine if the RAG contains a cycle and indicate it. (1 mark)

c) Deduce if there is a deadlock. Justify your answer based on the rule. (2 marks)

End of questions

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